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# United States Senate

4-5-16

## MEMORANDUM

**April 26, 1967**

**To:** **All Committee Members**

**From:** **James J. Gehrig**

**Attached is a summary of the Phillips Report provided the Committee by NASA. It is the same summary that was provided to the House Subcommittee on Oversight.**

April 15, 1967

MEMORANDUM for Hon. Olin Teague

SUBJECT: NASA's 1965 Management Review of North American Aviation

Early in the Apollo Command and Service Module and S-II Stage Programs, the performance requirements, key milestones, and cost plans were agreed upon by NASA and North American Aviation and formed the basis for negotiated cost plus fixed fee contracts between the government and North American Aviation, Inc. As the programs progressed, we experienced a trend of slippages in key milestone accomplishment, shortcomings in equipment performance and increasing costs.

By the fall of 1965 it was apparent that the rate of progress in both the CSM and S-II programs at the Space and Information Systems Division (S&ID) of North American Aviation, Inc. were not sufficient to meet the requirements of the Apollo Program. It appeared that both projects would diverge even further from planned technical, cost and schedule achievement despite the efforts of the two cognizant MSF Centers, unless immediate remedial action was taken. We therefore decided to undertake a high level review of the contractor's operations with a view to reexamining the reasons for past and current difficulties so that necessary corrective actions could be identified and taken to improve program position as rapidly as possible and to provide greater assurance that future technical, cost and schedule goals would be met or bettered.

The scope of the review included an examination of the NAA corporate organization and its relationship to and influence on the activities of the Space and Information Systems Division, the operating Division charged with the execution of the CSM and S-II Programs. The internal activities and management effectiveness of the Space and Information Systems Division were examined in depth in the following areas: Program Planning and Control; Contracting, Pricing, Subcontracting, Purchasing; Engineering; Manufacturing; and Reliability and Quality Assurance. The Review also included examination of North American off-site program activities in support of the Apollo Program at the Kennedy Space Center and at the Mississippi Test Facility.

The members of the NASA Review Team were specifically chosen for their competence in large R&D program management, their experience with S&ID and their intimate knowledge of the CSM and S-II Programs. The findings, therefore, comprised the views of knowledgeable and responsible government personnel directly involved with these programs.

The Review was conducted in the late fall of 1965. On December 19, 1965, a verbal presentation was given by the Review Team to the NASA Associate Administrator for Manned Space Flight and to North American Corporate President and staff and the management of their Space and Information Division. Following the presentation, a comprehensive set of notes describing the Review Team's findings in detail, and recommendations for improvement were provided to the contractor for appropriate action.

Between December 1965 and April 1966, North American organized an extensive effort headed by Mr. Atwood, NAA Corporate President, to review the NASA findings in detail and take appropriate action. An interim review was held for Mr. Atwood and Dr. Mueller in January 1966. The NASA Review Team reassembled at the contractor's plant April 22-27, 1966 to evaluate NAA progress in correcting deficiencies found by NASA and to assess the current situation and outlook. The contractor described the corrective actions already taken and outlined planned actions for further improvement, many of which were in the process of being implemented.

We concluded our Review Team meetings with North American on April 28, 1966, when we presented to the contractor and discussed our assessment of the actions taken and planned, and identified the areas in which we considered greater effort was still required. At this time we determined that the contractor had made substantial progress to correct many of the deficiencies noted, and that the planned future course of action provided confidence that the required rate of improvement in the posture of the CSM and S-II Programs would be achieved. We decided, therefore, to continue our activities with the contractor on these improvement efforts through our Center program management organizations in conjunction with their assigned responsibilities for the management of these contracts.

The most significant findings of our team a year ago last December, corrective actions taken by NAA, and our assessment of the situation at the conclusion of the Review Team activities in April 1966 are as follows:

Program Planning and Control

We found several inadequacies in this area. Work tasks were inadequately defined and scheduled, there were several different rather than a single integrated set of program and operating schedules and, largely as a result of these conditions, budget and cost control were deficient. There was a lack of visibility as to detailed status of the CSM and S-II programs.

One of the actions taken by NAA was to establish a disciplined, integrated planning and control system throughout S&ID. An important element was a system of work package management which not only defines project work and individual responsibility for that work but also provides a much improved basis for cost estimating and cost control. Concurrently, improved methods of providing each level of management with an assessment of program status, problems and outlook were implemented to enable NAA to better measure the progress of work against the plans and to identify problems and act on them in a timely manner.

Contracting, Pricing, Subcontracting and Purchasing

Many of the deficiencies we found in these areas trace back to the deficiencies in organization and program planning and control. Inadequate task definition, lack of integrated planning and control, poor status visibility and lack of strong program management organization all contributed to inaccurate cost estimates and lack of planning and disciplined follow through to insure prompt definitization of contracts and change orders. These problems were reflected in lengthy negotiation delays with both NASA and subcontractors.

The improvements in program planning and control and the strengthening of the program management organization provided a basis for improvement in contracting, pricing, subcontracting and purchasing. Standards and controls were improved in these areas within S&ID and we were beginning to see results in definitization of subcontracts and a significant increase in the effectiveness of the purchasing function.

The trend of improvement noted in the management of the CSM program confirmed the soundness of the incentive features of the CSM contract, which became effective in October 1965. A major problem still existed, however, regarding conversion of the S-II contract to an incentive basis. Continued efforts during 1965 to reach a program position where we could negotiate such an incentive contract were unsuccessful.

Engineering

We found four major deficiencies in S&ID engineering. The way in which work was decided between the central engineering organization and the program managers organization caused inefficiencies and made it difficult to identify and place accountability for required engineering outputs. In the case of the CSM, for example, engineering planning did exist at the lower levels of the organization but there was inadequate visibility and control of the work of the many engineering groups by higher levels of supervision and management. As a result, sufficient engineering effort was not being applied to the Block II spacecraft design, and Block II engineering releases and program schedule fell behind. An inadequate system engineering job was being done in the process of developing engineering design releases from NASA stage requirements. The engineering planning and output was not supporting program schedules. Procedures for configuration management, as agreed to by NAA and NASA were not being adhered to by the engineering organizations.

The consolidation of the engineering effort to more directly respond to the Program Managers' requirements and the improvement in defining and scheduling engineering tasks had significantly improved the timeliness and quality of the engineering for both the CSM and S-II.

The better identification of engineering tasks and more direct control of S&ID engineering resources by the CSM and S-II Program Manager strengthened project engineering and enabled these managers to do a better job of systems engineering by being in an improved position to control the individual design tasks to meet the system performance requirements stated by NASA. Significant progress was noted toward meeting the configuration management requirements established by NASA through more disciplined use of configuration control boards for decision making on engineering changes.

Manufacturing and Quality

These two areas will be discussed together since they are so related in terms of the quality of manufacturing workmanship and the effectiveness of the disciplines established to get the required level of quality in the delivered product.

Problems that resulted in rejection and rework were being experienced in the S-II program in manufacturing processes for bonding and welding, similar to earlier problems in spacecraft airframe fabrication. These difficulties were still being experienced in S-II in April 1966 although some improvements were noted.

Several of the problems we found in manufacturing stemmed from the lack of proper support from engineering. Late or incomplete engineering, for example had a direct impact on manufacturing's ability to meet their commitments to the program. The improvements made in engineering including more timely releases and engineering support on the factory floor, resulted in increased effectiveness in manufacturing.

NAA had implemented stronger material control and parts expediting and manufacturing supervision had improved. These actions contributed to the general efficiency of the work force.

In December 1965 Quality was not meeting NASA standards. This was evidenced by the large number of discrepencies discovered by both NAA and NASA inspectors and "corrections" type engineering orders issued by engineering. We found that quality trend data was not being used effectively by NAA management to measure performance and pinpoint and correct quality problems at the sources.

A major organizations change which provided stronger Quality organization support to the manufacturing and test organization and to the CSM and S-II program managers had been accomplished by April 1966. Program management attention to quality trend data, as a measure of performance, had increased.

The net effect of all these actions by NAA was improved adherence to Quality Assurance disciplines, and improving workmanship as evidence by a downward trend, from December 1965 to April 1966, in the discrepency rate.

To summarize, in each of these problem areas we identified in December 1965, we found in degrees varying from fair to excellent, improvement in the situation in April 1966. I do not intend, however, to leave the impression that we were completely satisfied with North American's performance. We did determine, however, that the corrective actions already taken and those in the process of being implemented in April 1966 were appropriate ones that reflected a responsive reaction by NAA to the criticism we had expressed during our review and provided some measure of confidence for improved future performance on the CSM and S-II Programs.

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
WASHINGTON, D.C. 20546

IN REPLY REFER TO: C:RLC:jf

4-25-16

April 25, 1967

Mr. James Gehrig
Staff Director
Committee on Aeronautical
and Space Sciences
United States Senate
Washington, D.C. 20510

Dear Jim:

Last evening we had a call from Miss Evangelist in Senator Brooks office who asked for the summary of the Phillips Report which we had provided to the House Subcommittee on NASA Oversight. This was mailed to Senator Brooks last night. Enclosed is a copy of the summary report for your information.

Sincerely yours,

Richard L. Callaghan
Assistant Administrator
for Legislative Affairs

Enclosure

(given to J.S.)

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